

QING YAN

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RESEARCH INTERESTS

Robotics, control algorithms, human-robot interaction, and mental models for embodied agents. Particular interest in using vision-language models and reinforcement learning to enhance pre-trained robot policies, and in modeling how robots should represent context when delegating tasks to people.

EDUCATION

Johns Hopkins University

Aug 2026 (expected)

M.S.E. in Robotics · Baltimore, MD

- Research focus: robotics, control algorithms, human-robot interaction, mental models

University of Washington

Jun 2024

B.S. in Electrical & Computer Engineering · Seattle, WA

- Cumulative GPA: 3.68 · Dean's List every quarter
- Selected coursework: Signal Processing, Robotics and Control System Analysis, Digital Circuits and Systems, Data Structures and Algorithms, Circuits, Calculus

PUBLICATIONS

- Gao, M., **Yan, Q.**, & Li, X. (2026). *GUIDES: Guidance Using Instructor-Distilled Embeddings for Pre-trained Robot Policy Enhancement*. IEEE International Conference on Robotics and Automation (ICRA), 2026. arxiv.org/abs/2511.03400 Advised by Prof. Chien-Ming Huang (JHU) and Prof. Jiachen Li (UC Riverside).
- Yang, W., & **Yan, Q.** (2026). *Scouting by Reward: VLM-to-IRL-Driven Player Selection for Esports*. MIT Sloan Sports Analytics Conference (under review). arxiv.org/abs/2604.14474 Advised by Prof. Anton Dahbura (JHU).

RESEARCH EXPERIENCE

Intuitive Computing Lab (LCSR), Johns Hopkins University

Sep 2024 – Present

Graduate Researcher, Human-Robot Interaction · Advised by Prof. Chien-Ming Huang

- Built **WeakRobotRestaurant**, a 2D human-robot interaction game in which limited-capability service robots ask people for help; logged contextual factors (workload, urgency, effort) and applied causal inference to isolate the effect of persuasion strategies on task acceptance.
- Framed the learned context-to-outcome mapping as a proto-mental world model for social delegation, specifying how an embodied robot should internally represent workload and urgency when choosing persuasion strategies.
- Developed a framework using VLM semantic guidance to upgrade traditional pre-trained robot policies (e.g., behavior cloning); validated on a physical UR5 robot arm. Results appear in publication 1.
- Designed the interactive infrastructure and an LLM-based voice command system for the Stretch RE1 mobile manipulator.

Sports Analytics Research Group, Johns Hopkins University

Sep 2025 – Jan 2026

Independent Researcher · Advised by Prof. Anton Dahbura

- Engineered an esports scouting system combining VLM and metadata embeddings with inverse reinforcement learning to identify professional players most similar to a target profile; validated with the esports organization Fnatic. Results appear in publication 2.

Sensor, Energy & Automation Laboratory (SEAL), University of Washington

Apr 2023 – Apr

2024

Embedded Team Associate

- Researched the Embedded Antenna Signals project, designing the signal-receiving module for the Heimdall ACU antenna system to detect weak electromagnetic interference using a transformer, preamplifier, and DAQ.
- Contributed to PCB simulation and noise-reduction experiments, and supported additional embedded hardware projects (Fruit Ripeness Sensor, Ammonia Sensor, Weak Signal Instrument).

INDUSTRY EXPERIENCE

KeploreAI

May 2025 – Present

Cofounder & COO · Santa Clara, CA

- Cofounded the company and its flagship product **Kepilot**, an AI-driven coding tool built on OpenDevIn with environment isolation and Docker integration to accelerate academic research and industrial R&D.
- Secured active Kepilot deployments and subscriptions across top research labs including Stanford, MIT, UC Berkeley, and UCLA.
- Introduced **Physmind**, a second-generation skills library providing end-to-end R&D solutions for magnetic-material, automotive, and robotics partners (e.g., Tianji Kirin, Geely Auto).
- Closed pre-seed funding and opened an active seed round; chief scientist is UCLA tenured Prof. Lin F. Yang (CS/EE).

Jianghuai Advance Technology Center (Qiyuan Lab)

Jul 2024 – Aug 2024

Control Algorithm Intern · Hefei, Anhui, China

- Trained the Qijiang No.1 humanoid robot for straight-leg walking using Proximal Policy Optimization (PPO) reinforcement learning, achieving fluent distance walking in simulation.
- Implemented a Universal Virtual Component (UVC) controller and tuned its parameters to enable stable bent-leg walking, transitioning the control strategy from manual parameter tuning to learning-based methods.

iFlytek Co., Ltd.

Jun 2022 – Sep 2022

Technology Department Intern

- Analyzed intelligent-manufacturing needs across Anhui Province through enterprise surveys, requirement analysis, and statistical reporting, with a focus on control systems and robotics.

SELECTED PROJECTS

Autonomous Self-Driving Wheelchair

ENGINE Capstone, UW

- Built an autonomous electric wheelchair (sponsored by Cyberwork Robotics) using a RealSense camera, RPLidar, and IMU. Combined SLAM and YOLO with ROS packages for navigation and obstacle avoidance, integrated the partner's Skylane lane-keeping system, and added an OGMD module to detect abnormal occupancy-grid features and switch navigation modes.

TECHNICAL SKILLS

Programming: Python, C++, Java, RobotC, Swift

AI & Algorithms: ROS, Computer Vision, Machine Learning / Reinforcement Learning, LLM / VLM, SLAM, Causal Inference, ReAct, OpenHands

Hardware & Platforms: Humanoid robots, Stretch, XLeRobot, autonomous wheelchair, UR5 arm, LiDAR, Intel RealSense, PCB design, Arduino, Raspberry Pi, FPGA

HONORS & AWARDS

- Dean's List, University of Washington (every quarter)
- Semifinalist, VEX Robotics World Championship (2019)
- Gold Award, VEX Robotics China National Championship (2019)
- Gold Award, VEX Robotics Asian Open (2018)
- First Prize (×2), China Adolescent Robotics Competition